RAVI SHAH
220102078

EE 632

Assignment 2

1. For the NOMA system model discussed in class with two uplink and two downlink users, taking interference into account, we have derived the outage probability (OP) for the far user (downlink scenario).

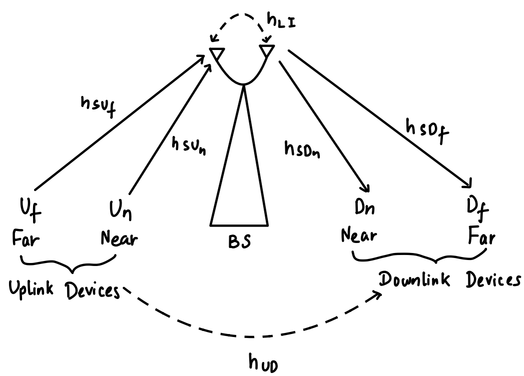
   (a) Derive the OP for the near user case (downlink scenario) and obtain the closed-form expression.

   (b) Perform Monte-Carlo simulations to obtain the OP of the near user. Compare the simulation results with the derived analytical expression for the near user.

   (c) Perform Monte-Carlo simulations to obtain the OP of the far user. Compare the simulation results with the analytical expression (derived in class) for the far user. Plot the OPs of (b) and (c) as a function of the transmit power values in the same figure.

   (d) Perform Monte-Carlo simulations to obtain the OPs of the near and far users in the uplink scenario. Comment on the OP results by comparing the downlink and uplink scenarios.

solⁿ. The case is,



Uplink channels

h_{SU_f} : $U_f \rightarrow S$

h_{SU_n} : $U_n \rightarrow S$

Downlink Channels

h_{SD_n} : $S \rightarrow D_n$

h_{SD_f} : $S \rightarrow D_f$

Interference for D_n,

h_{fn} : $U_f \rightarrow D_n$ $\qquad$ h_{nn} : $U_n \rightarrow D_n$

Let P_s be the BS power budget and a_f, a_n be the far and near user power allocation coefficients respectively.

Let P_{Uf}, P_{Un} be the uplink far and near user transmit power respectively.

Let N_0 be the noise variance for AWGN.

If x_f and x_n are the expected signals at far and near user respectively then the SINR at Dn with SIC is,

$$\gamma_{D_n}^{x_f} = \frac{a_f |h_{SD_n}|^2 \frac{P_s}{N_0}}{a_n |h_{SD_n}|^2 \frac{P_s}{N_0} + |h_{fn}|^2 \frac{P_{Uf}}{N_0} + |h_{nn}|^2 \frac{P_{Un}}{N_0} + 1} \quad , \quad \gamma_{D_n}^{x_n} = \frac{a_n |h_{SD_n}|^2 \frac{P_s}{N_0}}{|h_{fn}|^2 \frac{P_{Uf}}{N_0} + |h_{nn}|^2 \frac{P_{Un}}{N_0} + 1}$$

Let $\gamma_{thf}^{DL}, \gamma_{thn}^{DL}$ be the SINR thresholds for x_f and x_n respectively.

$\therefore$ The outage probability (OP) of Dn is

$$P_{out}^{D_n} = 1 - \Pr\left\{ \gamma_{D_n}^{x_f} > \gamma_{thf}^{DL} \, , \, \gamma_{D_n}^{x_n} > \gamma_{thn}^{DL} \right\}$$

Consider $\Pr\left\{ \gamma_{D_n}^{x_f} > \gamma_{thf}^{DL} \, , \, \gamma_{D_n}^{x_n} > \gamma_{thn}^{DL} \right\}$,

$$\frac{a_f |h_{SD_n}|^2 \frac{P_s}{N_0}}{a_n |h_{SD_n}|^2 \frac{P_s}{N_0} + |h_{fn}|^2 \frac{P_{Uf}}{N_0} + |h_{nn}|^2 \frac{P_{Un}}{N_0} + 1} > \gamma_{thf}^{DL}$$

and

$$\frac{a_n |h_{SD_n}|^2 \frac{P_s}{N_0}}{|h_{fn}|^2 \frac{P_{Uf}}{N_0} + |h_{nn}|^2 \frac{P_{Un}}{N_0} + 1} > \gamma_{thn}^{DL}$$

Let the RVs be $Z = |h_{SD_n}|^2 \sim$ Exponential with $\lambda = \frac{1}{\beta_{SD_n}}$

$Y = |h_{fn}|^2 \sim$ Exponential with $\lambda = \frac{1}{\beta_{fn}}$

$X = |h_{nn}|^2 \sim$ Exponential with $\lambda = \frac{1}{\beta_{nn}}$

Also, let $\gamma_s = \frac{P_s}{N_0}$, $\gamma_{Uf} = \frac{P_{Uf}}{N_0}$, $\gamma_{Un} = \frac{P_{Un}}{N_0}$

So we have,

$$\frac{a_f Z \gamma_s}{a_n Z \gamma_s + Y \gamma_{uf} + X \gamma_{un} + 1} > \gamma_{th_f}^{DL}$$

and

$$\frac{a_n Z \gamma_s}{Y \gamma_{uf} + X \gamma_{un} + 1} > \gamma_{th_n}^{DL}$$

Or,

$$Z > \frac{\gamma_{th_f}^{DL}}{(a_f - a_n \gamma_{th_f}^{DL})\gamma_s} \left(Y \gamma_{uf} + X \gamma_{un} + 1\right)$$

and

$$Z > \frac{\gamma_{th_n}^{DL}}{a_n \gamma_s} \left(Y \gamma_{uf} + X \gamma_{un} + 1\right)$$

Let $\phi = \max \left\{ \frac{\gamma_{th_f}^{DL}}{(a_f - a_n \gamma_{th_f}^{DL})\gamma_s} , \frac{\gamma_{th_n}^{DL}}{a_n \gamma_s} \right\}$

Then $Z > \phi \left(Y \gamma_{uf} + X \gamma_{un} + 1 \right)$

So, $P_{out}^{Dn} = 1 - P_r \left\{ Z > \phi \left(Y \gamma_{uf} + X \gamma_{un} + 1 \right) \right\}$

$$= 1 - \int_{\phi(\gamma_{uf} y + \gamma_{un} x + 1)}^{0} f_Z(z) \int_0^{\infty} f_Y(y) \int_0^{\infty} f_X(x) \, dx \, dy \, dz$$

Using exponential CDF,

$$P_{out}^{Dn} = 1 - \left\{ e^{-\frac{(\gamma_{uf} y + \gamma_{un} x + 1)\phi}{\beta_{SDn}}} \int_0^\infty f_Y(y) \int_0^\infty f_X(x)\, dx\, dy \right\}$$

$$= 1 - e^{-\frac{\phi}{\beta_{SDn}}} \int_0^\infty e^{-\frac{\gamma_{uf} y}{\beta_{SDn}}\phi} \cdot \frac{1}{\beta_{fn}} e^{-\frac{y}{\beta_{fn}}} dy \int_0^\infty e^{-\frac{\gamma_{uf} x}{\beta_{SDn}}\phi} \cdot \frac{1}{\beta_{nn}} e^{-\frac{x}{\beta_{nn}}} dx$$

$$= 1 - e^{-\frac{\phi}{\beta_{SDn}}} \cdot \left(\frac{\frac{1}{\beta_{fn}}}{\frac{1}{\beta_{fn}} + \frac{\phi \gamma_{uf}}{\beta_{SDn}}} \right) \cdot \left(\frac{\frac{1}{\beta_{nn}}}{\frac{1}{\beta_{nn}} + \frac{\phi \gamma_{un}}{\beta_{SDn}}} \right)$$

Or

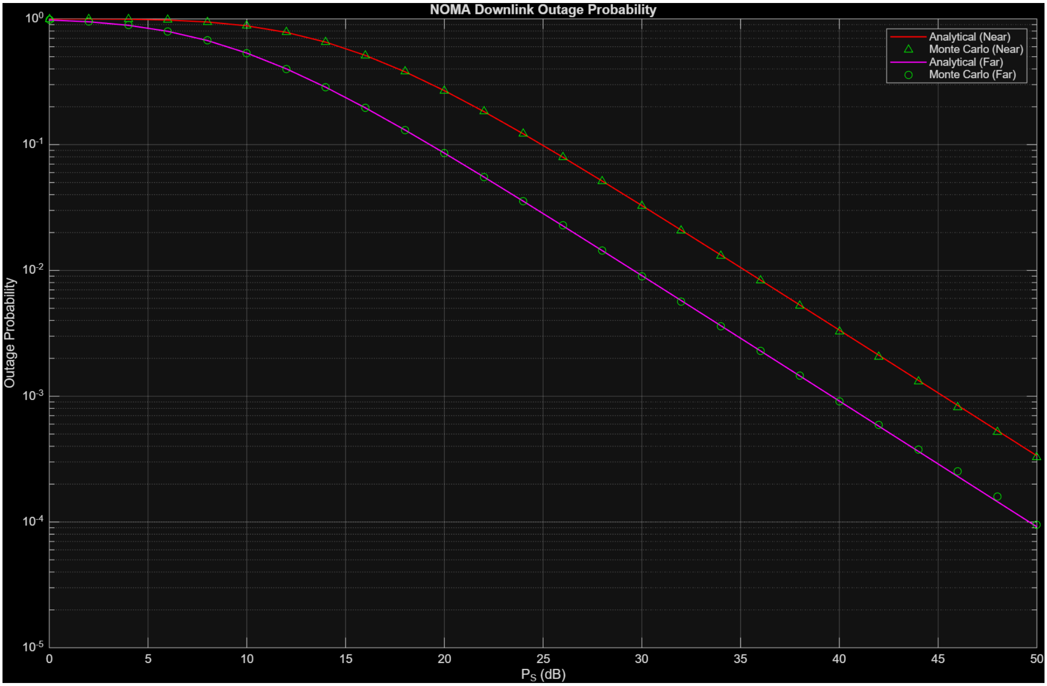
$$P_{out}^{Dn} = 1 - \frac{\beta_{SDn}^2}{(\beta_{SDn} + \phi \gamma_{uf} \beta_{fn})(\beta_{SDn} + \phi \gamma_{un} \beta_{nn})} e^{-\frac{\phi}{\beta_{SDn}}}$$

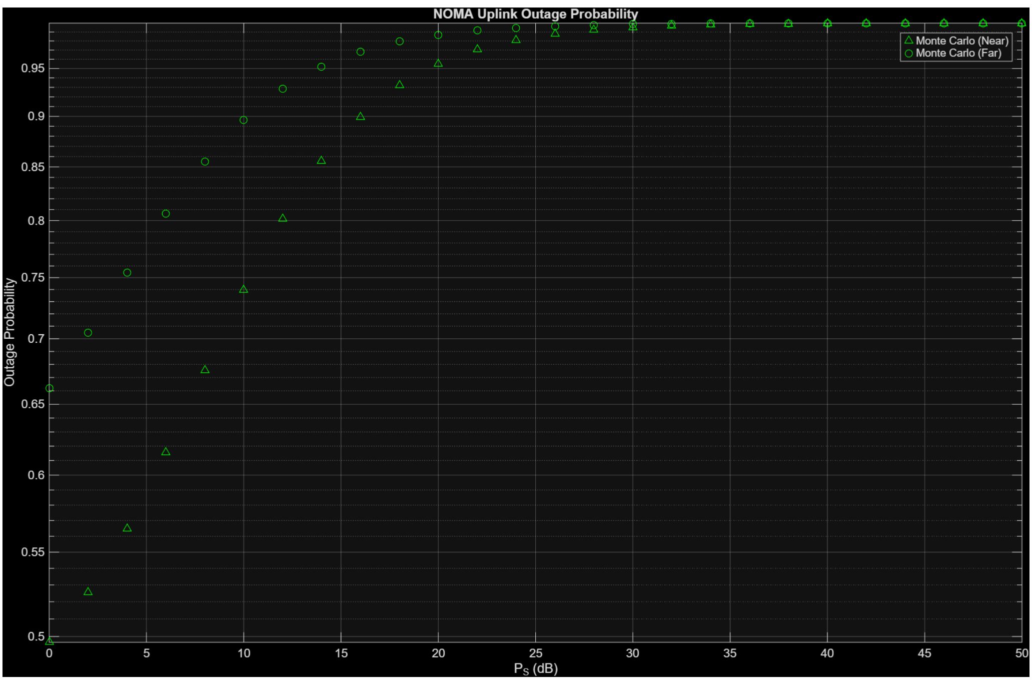
with,

$$\phi = \max \left\{ \frac{\gamma_{thf}^{DL}}{(a_f - a_n \gamma_{thf}^{DL})\gamma_s}, \frac{\gamma_{thn}^{DL}}{a_n \gamma_s} \right\}$$

Using simulations we can see that the simulated outage probabilities closely follow the analytical expression.

In downlink scenario, OPs are decreasing with increasing transmit power P_s because the SINRs increases whereas in the uplink scenario the opposite is observed as now the loop interference decreases the SINRs with increasing P_s.


NOMA Downlink Outage Probability
Analytical (Near)
Monte Carlo (Near)
Analytical (Far)
Monte Carlo (Far)
Outage Probability
P_S (dB)


NOMA Uplink Outage Probability
Monte Carlo (Near)
Monte Carlo (Far)
Outage Probability
P_S (dB)

3. Consider the joint sensing and communication (JSAC) system model with a passive sensing receiver as shown in Fig. 2. In class, we have discussed the detection versus communication tradeoff for this system model.

   We formulated the optimization problem to optimally allocate power to the sensing target and communication user, such that the detection

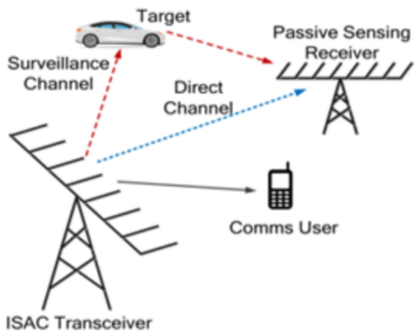


Figure 2: Joint passive sensing and communication.

probability is maximized while ensuring a minimum communication rate as given by

$$\begin{aligned} \max_{P_R, P_C} \quad & \mathcal{P}_D && (1) \\ \text{s.t.} \quad & R \geq R_{th}, \\ & P_R + P_C = P_T, \end{aligned}$$

where P_R and P_C represent the transmit power of the radar and communication signals, respectively, and P_T is the total power budget. $\mathcal{P}_D$ denotes the radar detection probability, and $R = \log(1 + P_C\gamma_c)$ is the achievable rate, with γ_c being the communication channel gain normalized by the noise variance. The rate threshold is denoted by R_{th}.

By sampling the received signals as L time-domain samples, the detection problem can be modeled as the binary hypothesis testing problem (ignoring clutter), which is discussed in class and given in notes as well.

(a) Find the detection probability $\mathcal{P}_D$ using a generalized likelihood ratio test (GLRT).

(b) Obtain the approximate $\mathcal{P}_D$ for the case with high direct-path SNR.

sol^{n}. We can model the binary hypothesis test as,

Target absent, H_0 : $\begin{cases} \underline{y_d} = \underline{\gamma_d}\, \underline{G_d}\, \underline{s_R} + \underline{n_d} \\ \underline{y_s} = \underline{n_s} \end{cases}$

Target present, H_1 : $\begin{array}{l} \underline{y_d} = \underline{\gamma_d}\, \underline{G_d}\, \underline{s_R} + \underline{n_d} \\ \underline{y_s} = \underline{\gamma_s}\, \underline{G_s}\, \underline{s_R} + \underline{n_s} \end{array}$

where

d : Direct channel

s : Surveillance channel

$\underset{L\times 1}{\underline{y_d}}$: Received direct channel L-sample sequence

$\underset{L\times 1}{\underline{y_s}}$: Received surveillance channel L-sample sequence

$\underset{L\times 1}{\underline{\gamma_d}}$: Gain of direct channel

$\underset{L\times 1}{\underline{\gamma_s}}$: Gain of surveillance channel

$\underset{L\times L}{\underline{G_d}}$: Delay - Doppler matrix of direct channel

$\underset{L\times L}{\underline{G_s}}$: Delay - Doppler matrix of surveillance channel

$\underset{L\times 1}{\underline{n_d}}$: AWGN present in direct channel

$\underset{L\times 1}{\underline{n_s}}$: AWGN present in surveillance channel

To solve this, the system compares the standard likelihood ratio against a certain threshold γ. The standard likelihood ratio function $L_{RB}(\underline{y_d}, \underline{y_s})$ is the ratio of joint PDF generated under each hypothesis

$$L_{RB}(\underline{y_d}, \underline{y_s}) = \frac{f((\underline{y_d}, \underline{y_s}) \cdot H_1)}{f((\underline{y_d}, \underline{y_s}) : H_0)}$$

Because parameters γ_d, γ_s and $\underline{s_R}$ are unknown, they are substituted with their Maximum Likelihood Estimations (MLE), denoted as $\hat{\theta}_1$ and $\hat{\theta}_0$. This creates the Generalized Likelihood Ratio Test (GLRT) function,

$$L_{GLRT}(\underline{y_d}, \underline{y_s}) = \frac{f((\underline{y_d}, \underline{y_s}) : \hat{\theta}_1, H_1)}{f((\underline{y_d}, \underline{y_s}) : \hat{\theta}_0, H_0)} \gtrless \gamma$$

MLE under each hypothesis

Under H_0, only $\underline{y_d}$ contains signal

$$\text{MLE of } \underline{s_R} : \quad \min_{\underline{s_R}} \| \underline{y_d} - \underline{\gamma_d}\, \underline{G_d}\, \underline{s_R} \|^2$$

$$\Rightarrow \hat{\underline{s_R}} = (\underline{G_d}^H \underline{G_d})^{-1} \underline{G_d}^H \underline{y_d}$$

Under H_1, since direct path is much stronger, similar to H_0

$$\hat{\underline{s_R}} = (\underline{G_d}^H \underline{G_d})^{-1} \underline{G_d}^H \underline{y_d}$$

$$\hat{\underline{s_R}} \propto \underline{y_d}$$

Now

$$L_{GLRT}(\underline{y_d}, \underline{y_s}) = \exp\left(-\frac{1}{\sigma^2}\left[\min J_1 - \min J_0\right]\right)$$

where,

$$J_0 = \| \underline{y_d} - \underline{\gamma_d}\, \underline{G_d}\, \underline{s_R} \|^2 + \| \underline{y_s} \|^2$$

$$J_1 = \| \underline{y_d} - \underline{\gamma_d}\, \underline{G_d}\, \underline{s_R} \|^2 + \| \underline{y_s} - \underline{\gamma_s}\, \underline{G_d}\, \underline{s_R} \|^2$$

Using $\hat{s}_R \propto y_d$

$$\underline{\gamma_s}\, \underline{G_s}\, \underline{s_R} \propto \underline{y_d}$$

or

$$\underline{y_s} \approx \alpha\, \underline{y_d} + \underline{n_s}$$

Now, the problem becomes

$$\min_{\alpha} \|\underline{y}_s - \alpha \underline{y}_d\|^2$$

This gives

$$\hat{\alpha} = \frac{\underline{y}_d^H \underline{y}_s}{\|\underline{y}_d\|^2}$$

So,

$$J_0 - J_1 = \|\underline{y}_s\|^2 - \left(\|\underline{y}_s\|^2 - \frac{|\underline{y}_d^H \underline{y}_s|^2}{\|\underline{y}_d\|^2}\right)$$

$$= \frac{|\underline{y}_d^H \underline{y}_s|^2}{\|\underline{y}_d\|^2}$$

$\therefore$ The final test statistic is

$$T = \frac{|\underline{y}_d^H \underline{y}_s|^2}{\|\underline{y}_d\|^2} \gtrless \gamma$$

Now, under H_0

$$T \sim \chi_2^2$$

under H_1

$$T \sim \chi_2^2(\lambda)$$

with $$\lambda = \frac{|\underline{y}_d^H(\underline{\gamma}_s \underline{G}_s \underline{s}_R)|^2}{\sigma^2 \|\underline{y}_d\|^2}$$

Since $\underline{y}_d \approx \underline{\gamma}_d \underline{G}_d \underline{s}_R$

$$\lambda = \frac{P_R \|\underline{\gamma}_d\|^2}{\sigma^2}$$

Now, $$R = \log_2(1 + P_c \gamma_c) \geq R_{th}$$

$$\Rightarrow P_c = \frac{2^{R_{th}} - 1}{\gamma_c}$$

Now, $R = \log_2 (1 + P_c \gamma_c) \geq R_{th}$

$$\Rightarrow P_c = \frac{2^{R_{th}} - 1}{\gamma_c}$$

Using $P_c + P_R = P_T$

$$\Rightarrow P_R = P_T - \frac{2R^{th} - 1}{\gamma_c}$$

Now, the standard result in GLRT is,

$$P_D = Q_1\left(\sqrt{2\lambda}, \sqrt{2\gamma}\right)$$

$\left\{ Q_1 : \text{First order Marcum-Q function} \right\}$

For our case,

non-centrality parameter $\lambda = \dfrac{P_R \|\underline{\gamma_d}\|^2}{\sigma^2} = \left(P_T - \dfrac{2^{R_{th}} - 1}{\gamma_c}\right)\dfrac{\|\underline{\gamma_d}\|^2}{\sigma^2}$

$$\therefore \quad P_D = Q_1\left(\sqrt{2\left(P_T - \frac{2^{R_{th}} - 1}{\gamma_c}\right)\frac{\|\underline{\gamma_d}\|^2}{\sigma^2}}, \sqrt{2\gamma}\right)$$